

Vaccine trials with adaptive designs: Promise or peril?

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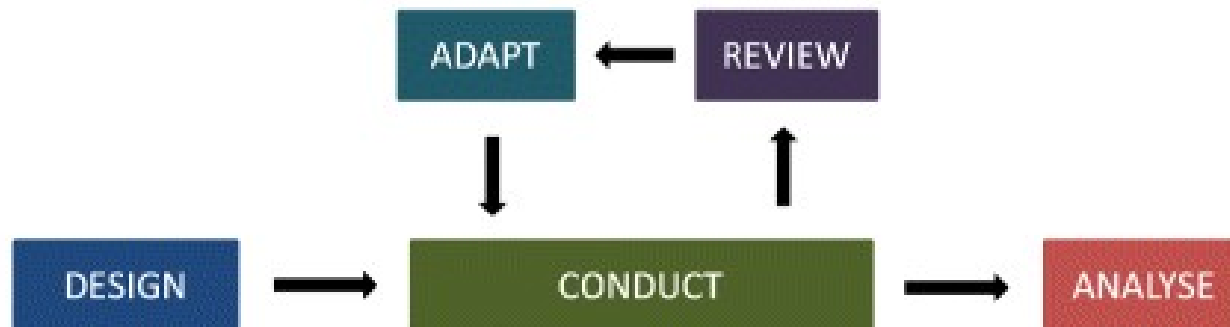


Conflict of Interest

- Co-authors James McMahon (**JM**), Peter Richmond (**PR**) and Tom Snelling (**TS**)
 - Membership of immunisation committees
 - Australian Technical Advisory Group on Immunisation (**PR and TS**)
 - Western Australia Vaccine Safety Advisory Committee (**PR**)
 - Member of vaccine scientific advisory boards (no personal remuneration)
 - GSK, Pfizer, CSL, Sanofi, AstraZeneca, Merck, Moderna and Novavax (**PR**)
 - Vaccine research
 - Investigator for industry sponsored multi-centre studies for vaccines (**JM and PR**)
 - Travel support to present at scientific conferences from Sanofi, Pfizer, Baxter and GSK (**PR**)
 - Research funding for investigator-initiated studies from GSK, Merck and Novartis (**PR**)
- There are no other potential conflicts of interest to declare.

Clinical trial designs

- It is important to design clinical trials carefully
- **Fixed designs** do not change as the trial progresses
- **Adaptive designs** *may* change as the trial progresses
 - Potential design modifications are pre-specified and are based on accumulated data



Example Adaptations

Sample size re-estimation

Early stopping
(e.g., efficacy/futility)

Arm adding/dropping

Potential Advantages

Flexibility

Operational efficiency

Attractive to funders

Reduced sample sizes

Potential Disadvantages

Setup overheads

Ongoing resource use
(e.g., database/statistics)

Some vaccine trials with adaptive designs

Trial	Area	Adaptive Elements	Adaptation Outcome/s
ORVAC	Rotavirus	Early stopping	Immunological/Clinical
OPTIMUM	Pertussis	Early stopping	Clinical
Pfizer	COVID-19	Seamless multi-phase; early stopping	Safety/Immunological/Clinical
Moderna	COVID-19	Early stopping	Clinical
Novavax	COVID-19	Early stopping	Clinical
Oxford-AstraZeneca	COVID-19	Early stopping	Clinical
PICOB00	COVID-19	Arm adding/dropping; early stopping	Immunological
BOOST-IC	COVID-19	Arm adding/dropping; early stopping	Immunological

Experience with PICOB00 and BOOST-IC

- Immunocompetent and immunocompromised populations in Australia
- Randomised to Pfizer, Moderna or Novavax
- Precision based early stopping on short-term immunological outcome
- Flexibility to add/drop arms to evaluate new vaccines

Strengths	Challenges

Considerations for vaccine trials with adaptive designs

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